

SECR TELECOM POSITION DASHBOARD

System Architecture, Data Flows, and Refactored Forms Summary

SECR TELECOM

1. OVERVIEW

The **South East Central Railway (SECR) Divisional Telecom Position Dashboard** is a Next.js web application designed for telecom engineers to record, monitor, and update the status of telecom circuits, cable infrastructure, and surveillance systems across three railway divisions: **Bilaspur**, **Raipur**, and **Nagpur**.

2. TECHNOLOGY STACK

- **Core System:** Next.js 16 (React 19, Turbopack) & TypeScript
- **Styling & Design System:** Vanilla CSS for maximum performance, responsiveness, and premium dark/glassmorphic aesthetics
- **State Management:** React Hooks (`useState` , `useMemo` , `useRef` , `useEffect`) for real-time calculations and zero-latency view updates

3. APPLICATION FLOW & USER OPERATIONS

- **Division Filter Context:** Globally filters major sections, sections, and stations for consistent reporting.
- **Categorized Circuit Workspace Selection:** Selecting a circuit dynamically displays its dedicated entry form in the viewport.
- **Real-time failure duration calculation:** Calculations are performed on-the-fly using `useMemo` , outputting formatted durations automatically.
- **"All OK" Workflow Accelerator:** Allows one-click logging of 100% healthy status across all elements and automatically advances to the next circuit.

4. SUMMARY OF REFACTORED DEDICATED FORMS

Each circuit has been upgraded from generic layouts to customized forms designed for quick data entry and aligned side-by-side:

CATEGORY	CIRCUIT NAME	CUSTOM FEATURES & FORMS LAYOUT
Network & Internet	PRS/UTS	Consolidated UTS/PRS. Fields: System Type, Major Section/Section, Station, Nature of Fault (Equipment/Link), Failure & Rectification Times, auto-calculated Duration, Reason, and Remarks.
Network & Internet	Railnet / Internet	Removed redundant searchable location dropdown. Provides dedicated Divisional Maintenance and HQ Maintenance tab layouts.
Network & Internet	Wi-Fi	Dedicated form tracking: Location of faulty AP, Failure & Rectification Times, auto-calculated Duration, Reason of Failure, and Remarks.
CCTV	CCTV Monitoring	Dedicated fields: Major Section/Section, Total CCTV/Not working CCTV, Live Feed To War Room Failed status, Failure & Rectification Times, Duration, Reason, and Remarks.
CCTV	CCTV Maintenance	Mirror layout of CCTV Monitoring, dedicated to hardware servicing and disk checks.
Cable Infrastructure	Cable Cut (OFC & Quad)	Laid out in clean 2-column grid: Major Section/Section, Station/Location, Km.No., Cable Type, Cable Cut by Whom, Failure & Rectification Times, Duration, Reason, and Remarks.
Cable Infrastructure	Temporary Joints	Excludes Station/Location field. Tracks Section/Yard, Km.No., cable type, temporary joints count, rectified joints, TDC, Action Plan, and auto-calculates Balance Joints.
Cable Infrastructure	Low Insulation	Excludes yard/section text inputs. Pairs Major Section, Section, Station/Location, Km.No., Cable Type, Total Insulation Faults, Rectified Date & Time, Balance Insulation Faults, Action Plan/TDC, and Remarks.
Display System	CGDM	Tracks platform number (PF No.), number of faulty guidance boards, Failure Time, Rectification Time, Duration, Reason, and Remarks.
Display System	TIB	Tracks location of faulty TIB, number of faulty boards, Failure Time, Rectification Time, Duration, Reason, and Remarks.

CATEGORY	CIRCUIT NAME	CUSTOM FEATURES & FORMS LAYOUT
Testing & Maintenance	Walkie-Talkie Repairing	Excludes icmsEntryNo requirement. Tracks Defective/Repaired set counts, Sets Condemned, target condemnation dates, repair status, action taken, and repair faults.

5. PROJECT FILE MAP

- Main Component Logic: `src/app/page.tsx`
- Global CSS Styling: `src/app/globals.css`
- Tasks Registry: `task.md`
- Walkthrough History: `walkthrough.md`